

Topic 7: Multivariate Derivatives¹

MULTIVARIATE FUNCTIONS

Consider a production function with 3 inputs and 2 outputs:

$$F(y_1, y_2, y_3) = \begin{bmatrix} F^1(y_1, y_2, y_3) \\ F^2(y_1, y_2, y_3) \end{bmatrix} = \begin{bmatrix} y_1 y_2 y_3 \\ \frac{y_1^\alpha y_2^{1-\alpha}}{y_3} \end{bmatrix}.$$

In such a case, we say $F : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ to be a **multivariate** (real) function with 3 variables, and F^1, F^2 here are the component functions of F .

Note that there is a Greek letter α which is not a variable, but it still affects the function values. In such a case, we call α a **parameter**, where sometimes we write

$$F(y_1, y_2, y_3; \alpha).$$

Consider firstly $f : \mathbb{R} \rightarrow \mathbb{R}$. Apply the linearization we mentioned before:

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x) + ah}{h} = 0$$

where a is the derivative evaluated at $x \in I$ for some interval $I \subset \mathbb{R}$.

We can extend this observation to multivariate functions.

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TOTAL DIFFERENTIATION**Definition 1** (Total Differentiation)

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and f maps an open set $E \subset \mathbb{R}^n$ to $\mathbb{R}^m$.

We say f is **differentiable** at $x \in E \subset \mathbb{R}^n$ if:

$\forall x \in E, \quad \exists m \times n$ matrix $A(x)$ such that

$$\lim_{\|h\| \rightarrow 0} \frac{\|f(x+h) - f(x) - A(x)h\|}{\|h\|} = 0$$

where $h \in \mathbb{R}^n$.

In this case, we say the transformation $A(x)$ is the **total differentiation** of f , denoted as

$$Df(x) = A(x).$$

Example 1

(1) What is the dimension of the 0 in $\|h\| \rightarrow 0$? Why is $h \in \mathbb{R}^n$ but not $\mathbb{R}^m$ or other dimensions?

(2) Write out the definition of total differentiation when $m = 1$ and $n = 1$. Compare it with the univariate differentiation that we studied.

Total differentiation involves finding the differential of a function that depends on more than 1 variable (e.g., n variables here).

PARTIAL DIFFERENTIATION

We also define another type of differentiation: the partial differentiation.

Partial differentiation focuses on finding the derivative of a function with respect to one variable while holding the other variables constant. We call this derivative a partial derivative.

Definition 2 (Partial Derivative)

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and f maps an open set $E \subset \mathbb{R}^n$ to $\mathbb{R}^m$.

For each $i = 1, 2, \dots, n$, the **partial derivative** of f with respect to the i -th component of $x = (x_1, x_2, \dots, x_n)$ at a point $x \in E$ is defined as:

$$\lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x_1, \dots, x_i, \dots, x_n)}{h}$$

The **partial derivative** of f with respect to x_i is denoted as

$$D_i f(x) \text{ or } \frac{\partial f}{\partial x_i}(x) \text{ or } f_i(x).$$

Example 2

Consider $n = 3$ and $m = 2$. What is the dimension of $Df(x)$? What is the dimension of $D_1 f(x)$? How many partial derivatives are there?

The following theorem links total differentiation to partial derivatives.

Theorem 1

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and f maps an open set $E \subset \mathbb{R}^n$ to $\mathbb{R}^m$.

Then, f is differentiable at $x \in E$ if and only if each of the component function f^k is differentiable at x .

Furthermore,

$$Df(x) = \begin{bmatrix} D_1 f(x) & \cdots & D_n f(x) \end{bmatrix} = \begin{bmatrix} D_1 f^1(x) & \cdots & D_n f^1(x) \\ \vdots & \ddots & \vdots \\ D_1 f^m(x) & \cdots & D_n f^m(x) \end{bmatrix}.$$

We also call the following matrix the **Jacobian matrix**:

$$J(f) = \begin{bmatrix} D_1 f & \cdots & D_n f \end{bmatrix} = \begin{bmatrix} D_1 f^1 & \cdots & D_n f^1 \\ \vdots & \ddots & \vdots \\ D_1 f^m & \cdots & D_n f^m \end{bmatrix}$$

For simplicity, let's start with the case when $m = 1$. This is an example to see the difference between a total differentiation and a partial differentiation.

Example 3

Let $U : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a utility function of two goods: coffee c and coffee mate m .

Let $U(c, m) = \log(c) \times \log(m)$, where the consumption of coffee mate is proportional to the consumption of coffee, specifically $m = 0.05c$ (i.e., 1 ounce of coffee mate is consumed with every 20 ounces of coffee).

- (1) What is the dimension of the total differentiation $DU(c, m)$? Find $DU(c, m)$.
- (2) What is the economic meaning of $D_c U(c, m)$ (that is, $\frac{\partial U(c, m)}{\partial c}$)?
- (3) What is the economic meaning of $D_m U(c, m)$ (that is, $\frac{\partial U(c, m)}{\partial m}$)?

$DU(c, m) = [D_c U(c, m) \quad D_m U(c, m)]$. Here, $D_c U(c, m) = \frac{\log(m)}{c}$ is the marginal utility of coffee, that is, the additional utility gained from consuming one more unit of coffee while holding the quantity of coffee mate constant. $D_m U(c, m) = \frac{\log(c)}{m}$ represents the marginal utility of coffee mate, or the additional utility gained from consuming one more unit of coffee mate while holding the quantity of coffee constant.

You may think the proper "value" of one more unit of coffee should not be evaluated holding the quantity of coffee mate constant. After all, I should have increased coffee mate consumption by 0.05 unit too (because $m = 0.05c$). In fact, this idea is captured by the total differential. We will discuss it later.

Example 4

Find the total differentiation $DF(y_1, y_2, y_3)$ of

$$F(y_1, y_2, y_3) = \begin{bmatrix} F^1(y_1, y_2, y_3) \\ F^2(y_1, y_2, y_3) \end{bmatrix} = \begin{bmatrix} y_1 y_2 y_3 \\ \frac{y_1^\alpha y_2^{1-\alpha}}{y_3} \end{bmatrix}.$$

Using the definition above, we can find

$$\begin{aligned} DF(y_1, y_2, y_3) &= \begin{bmatrix} D_1 F^1(y_1, y_2, y_3) & D_2 F^1(y_1, y_2, y_3) & D_3 F^1(y_1, y_2, y_3) \\ D_1 F^2(y_1, y_2, y_3) & D_2 F^2(y_1, y_2, y_3) & D_3 F^2(y_1, y_2, y_3) \end{bmatrix} \\ &= \begin{bmatrix} y_2 y_3 & y_1 y_3 & y_1 y_2 \\ \alpha \left(\frac{y_2}{y_1}\right)^{1-\alpha} \left(\frac{1}{y_3}\right) & (1-\alpha) \left(\frac{y_1}{y_2}\right)^\alpha \left(\frac{1}{y_3}\right) & -\frac{y_1^\alpha y_2^{1-\alpha}}{y_3^2} \end{bmatrix} \end{aligned}$$

Sometimes we denote

$$\nabla f(x) = \begin{bmatrix} D_1 f(x) \\ \vdots \\ D_n f(x) \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x_1}(x) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x) \end{bmatrix}$$

as the **gradient** of f at x . Note that $Df(x)$ and $\nabla f(x)$ are effectively the same thing: They are just transposes of each others.

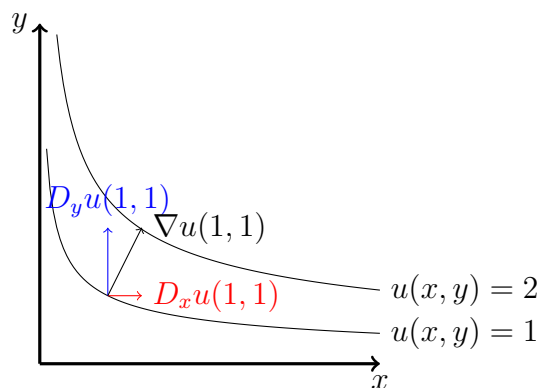
If f is real-valued (the codomain is $\mathbb{R}^1$ but not $\mathbb{R}^m$), then $\nabla f(x)$ is the **gradient vector** of f at x .

Gradient has an intuitive interpretation: it represents when we move along each direction, how much the function value will change. Take a Cobb-Douglas utility function $u(x, y) = x^{\frac{1}{3}} y^{\frac{2}{3}}$ as an example. Then we can find

$$\nabla u(x, y) = \begin{bmatrix} D_x u(x, y) \\ D_y u(x, y) \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \left(\frac{y}{x}\right)^{\frac{2}{3}} \\ \frac{2}{3} \left(\frac{x}{y}\right)^{\frac{1}{3}} \end{bmatrix}$$

Hence $D_x u(1, 1) = \frac{1}{3}$ and $D_y u(1, 1) = \frac{2}{3}$. That is, when we move from $(1, 1)$ *one marginal unit toward* x , it increases the utility by $\frac{1}{3}$ units; when we move from $(1, 1)$ *one marginal unit*

toward y , it increases the utility by $\frac{2}{3}$ units. And when we move one unit along both directions, the total utility level adjusted is determined by the composition from both directions.



From the graph above of the level curve of the utility function u , you may also observe another property of the gradient: the gradient of a function is *orthogonal* to the level curve. We will apply this fact when we talk about Lagrangian in constrained optimization.

TOTAL DIFFERENTIAL

The most important application of total differentiation is to determine the **total differential**.²

We learned earlier that

$$Df(x) = \begin{bmatrix} D_1 f(x) & \cdots & D_n f(x) \end{bmatrix} = \begin{bmatrix} D_1 f^1(x) & \cdots & D_n f^1(x) \\ \vdots & \ddots & \vdots \\ D_1 f^m(x) & \cdots & D_n f^m(x) \end{bmatrix}$$

where $x = (x_1, x_2, \dots, x_n)$.

In the univariate case, if we want to find how $f(x)$ responds to small changes in the input variable x , we can calculate $\frac{df(x)}{dx}$.

In the multivariate case, we may also want to find how $f(x)$ responds to small changes in

²The terms total differentiation, partial differentiation, and total differential can be very confusing. Make sure you know the definition of these three when you are reviewing them.

the input vector x . To do this, let's observe what would happen if we multiply $Df(x)$ by dx . Note that $dx = (dx_1, dx_2, \dots, dx_n)$.

If we exploit the notation:

$$df(x) \equiv Df(x)dx = D_1f(x)dx_1 + D_2f(x)dx_2 + \dots + D_nf(x)dx_n,$$

then we can write the change of f into the changes in each variable.

Moreover, when we want to find the comparative static with respect to x_1 only (but not the vector x), we can calculate

$$\frac{df(x)}{dx_1} = D_1f(x) + D_2f(x)\frac{dx_2}{dx_1} + \dots + D_nf(x)\frac{dx_n}{dx_1}.$$

In other words, when we are evaluating the effects on a small change in x_1 on $f(x)$, we need to consider not only the direct effect ($D_1f(x)$) but also the indirect effect of x_1 on other variables.

Example 5

Let $U : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a utility function of two goods: coffee c and coffee mate m .

Let $U(c, m) = \log(c) \times \log(m)$, where the consumption of coffee mate is proportional to the consumption of coffee, specifically $m = 0.05c$ (i.e., 1 ounce of coffee mate is consumed with every 20 ounces of coffee).

(1) Write out again the $D_cU(c, m)$ (that is, $\frac{\partial U(c, m)}{\partial c}$).

(2) Write out the formula of $\frac{dU(c, m)}{dc}$. What is the economic meaning of $\frac{dU(c, m)}{dc}$? What is the economic meaning of each term?

(2) Find $\frac{dU(c, m)}{dc}$. Compare it with $D_cU(c, m)$. Is one of them always greater than another (assume $c, m > 1$)? If so, why?

To find the total derivative $\frac{dU(c, m)}{dc}$, we must account for the fact that m depends on c through the relationship $m = 0.05c$:

$$\frac{dU(c, m)}{dc} = \frac{\partial U}{\partial c} \cdot \frac{dc}{dc} + \frac{\partial U}{\partial m} \cdot \frac{dm}{dc}$$

We already found in the previous question that:

$$\frac{\partial U}{\partial c} = \frac{\log(m)}{c}, \quad \text{and} \quad \frac{\partial U}{\partial m} = \frac{\log(c)}{m}$$

Also, since $m = 0.05c$:

$$\frac{dm}{dc} = 0.05$$

Thus, the total derivative is:

$$\frac{dU(c, m)}{dc} = \frac{\log(m)}{c} \cdot 1 + \frac{\log(c)}{m} \cdot 0.05$$

- The term $\frac{dU(c, m)}{dc}$ captures the overall rate of change in utility with respect to coffee, accounting for both the direct effect of consuming more coffee and the indirect effect through changes in the consumption of coffee mate.
- The term $\frac{\log(m)}{c}$ captures the direct impact of increasing c .
- The term $\frac{\log(c)}{m} \cdot 0.05$ captures the indirect effect due to the proportional increase in m with c .

The total derivative is greater (when $c, m > 1$) because it includes the indirect effect of consuming more coffee mate when having more coffee.

MULTIVARIATE CONVEX FUNCTION

Recall that we defined the convex function:

Definition 3 (Convex function)

Let $f : X \rightarrow \mathbb{R}$. We say f is a **convex function** if for any $x_0, x_1 \in X$, $\lambda \in [0, 1]$,

$$f(\lambda x_0 + (1 - \lambda)x_1) \leq \lambda f(x_0) + (1 - \lambda)f(x_1).$$

And we have studied that when $X = \mathbb{R}$, there is a way to use derivative to verify if a function is convex (if derivative exists).

Theorem 2

For a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is differentiable on an interval I , the the function f is convex if its first derivative is increasing on I . That is,

$$f'(x_1) \leq f'(x_2) \text{ whenever } x_1 < x_2 \in I$$

For a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is twice differentiable on an interval I , the the function f is convex if its second derivative is positive on I . That is,

$$f''(x) > 0 \quad \forall x \in I$$

How about $X = \mathbb{R}^n$? Is there a corresponding way to verify that a function is convex? The answer is yes. Analogous to the condition $f''(x) > 0$ in the univariate case, the verification of convexity depends on second order derivatives.

Definition 4

Let f is twice-differentiable. Then the **Hessian matrix** is defined as

$$H_f = J(\nabla f) = \begin{bmatrix} D_{11}f & \cdots & D_{1n}f \\ \vdots & \ddots & \vdots \\ D_{n1}f & \cdots & D_{nn}f \end{bmatrix} = \begin{bmatrix} \frac{\partial^2 f}{(\partial x_1)^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{(\partial x_n)^2} \end{bmatrix}.$$

Example 6

Let $U(c, m) = \log(c) \times \log(m)$, where the consumption of coffee mate is proportional to the consumption of coffee, specifically $m = 0.05c$ (i.e., 1 ounce of coffee mate is consumed with every 20 ounces of coffee).

Find the Hessian matrix for $U(c, m)$.

When taking partial derivatives, the order does not matter.

Theorem 3: Young's theorem

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be twice-differentiable. Then

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$$

for any $i, j = 1, \dots, n$.

Theorem 4

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable in an open set $X \subset \mathbb{R}^n$.

f is convex (concave) in X if and only if the Hessian matrix of f is *positive (negative) semi-definite* at every point $x \in X$.

Similarly, f is strictly convex (concave) in X if and only if the Hessian matrix of f is *positive (negative) definite* at every point $x \in X$.

Note:

1. A matrix is positive definite if all the leading principal minors are positive.
2. A matrix is positive semi-definite if all the leading principal minors are non-negative.
3. A matrix is negative definite if all the leading principal minors of the matrix alternate in sign, starting with a negative sign.
4. A matrix is negative semi-definite if all the odd-order principal minors are non-positive, and all the even-order principal minors are non-negative.

We have a similar second-order test for quasi-convex/concave functions. We first define the **bordered Hessian matrix** of a twice differentiable function f ,

$$\overline{H}_f = \begin{bmatrix} 0 & D_1 f & \cdots & D_n f \\ D_1 f & D_{11} f & \cdots & D_{1n} f \\ \vdots & \vdots & \ddots & \vdots \\ D_n f & D_{n1} f & \cdots & D_{nn} f \end{bmatrix}.$$

It is exact the Hessian matrix of f with a “border” of first derivatives.

Theorem 5

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable in an open set $X \subset \mathbb{R}^n$.

Let $\overline{H}$ be the bordered Hessian matrix of f and $\overline{H}_i$ be the i -th order leading principal minor of $\overline{H}$.

Then, $\forall x \in X$,

- f is quasi-convex if $\det(\overline{H}_i) \leq 0$ for $i \geq 2$.
- f is quasi-concave if $\det(\overline{H}_i) \leq 0$ for all even i and $\det(\overline{H}_i) \geq 0$ for all odd i for all odd $i \geq 3$.
- f is strictly quasi-convex if $\det(\overline{H}_i) < 0$ for $i \geq 2$.
- f is strictly quasi-concave if $\det(\overline{H}_i) < 0$ for all even i and $\det(\overline{H}_i) > 0$ for all odd i for all odd $i \geq 3$.

Example 7

Determine whether the following functions are convex/concave/quasi-convex/quasi-concave.

1. $f(x, y) = x^2 + y^2$
2. $g(x, y) = xy$

One can solve that

$$H_f = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \quad \text{and} \quad H_g = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

where H_f is positive semi-definite (in fact it is positive definite) and H_g is indefinite. Hence, f is convex (and not concave). g is neither convex nor concave.

The bordered Hessians of f and g are

$$\overline{H}_f = \begin{bmatrix} 0 & 2x & 2y \\ 2x & 2 & 0 \\ 2y & 0 & 2 \end{bmatrix} \quad \text{and} \quad \overline{H}_g = \begin{bmatrix} 0 & y & x \\ y & 0 & 1 \\ x & 1 & 0 \end{bmatrix},$$

and the determinants of the leading principal minors are

$$\begin{aligned}\det(\overline{H}_{f_2}) &= -4y^2 \leq 0, & \det(\overline{H}_{f_3}) &= -8x^2 - 8y^2 \leq 0, \\ \det(\overline{H}_{g_2}) &= -x^2 \leq 0, & \det(\overline{H}_{g_3}) &= 0.\end{aligned}$$

Hence both f and g are quasi-convex.